

DATA SCIENCE RESOURCES

HR & Employee Analytics Projects

Project 1. Employee Attrition Prediction (Churn Analysis)

Problem Statement: Employee turnover is costly and disrupts team productivity. HR teams struggle to predict which employees are at risk of leaving and why.

Objective: Build a machine learning model to predict whether an employee is likely to leave the organization, based on factors like job role, satisfaction, salary, and tenure.

Dataset Source: [IBM HR Analytics Employee Attrition Dataset](#)

Techniques & Concepts

- Binary classification
- Feature importance analysis (e.g., SHAP)
- Handling class imbalance (SMOTE, class weights)
- HR dashboard metrics (voluntary vs involuntary attrition)

Tools & Libraries: Python, Pandas, Scikit-learn, XGBoost, SHAP, Streamlit for deployment

Expected Outcome: A predictive model and dashboard to alert HR teams about employees at high risk of attrition and suggest intervention strategies.

Project 2. Resume Matching Algorithm for Automated Recruitment

Problem Statement: Recruiters spend hours screening resumes. Manual matching is inefficient and prone to bias, especially when hiring at scale.

Objective: Build a Natural Language Processing (NLP) based system to rank and match candidate resumes with job descriptions based on skill similarity.

Dataset Source: [Simulated or scraped job descriptions and resumes](#)

Techniques & Concepts

- Text preprocessing (tokenization, stopwords removal)
- TF-IDF and cosine similarity for matching
- Named Entity Recognition (skills, tools, degrees)
- Ranking system for top candidates

Tools & Libraries: Python, NLTK, SpaCy, Scikit-learn, Gensim, Flask or Streamlit

Expected Outcome: A matching system that takes a job description and ranks uploaded resumes by relevance, speeding up hiring processes.

Project 3. Workforce Diversity Analytics

Problem Statement: Organizations aim to improve diversity and inclusion, but they often lack data-driven insights into representation, bias, or inequality in hiring and promotion.

Objective: Analyze diversity across gender, ethnicity, department, and experience levels. Identify patterns of inequality and suggest interventions for improvement.

Dataset Source: [Simulated HRIS datasets](#)

Techniques & Concepts

- Descriptive analytics and EDA
- Cross-tabulation and group analysis (e.g., by gender/role)
- Pay gap analysis
- Data storytelling for HR insights

Tools & Libraries: Python, Pandas, Seaborn, Plotly, Tableau or Power BI

Expected Outcome: Interactive dashboards and reports showing diversity stats and recommendations, supporting HR and DEI (Diversity, Equity, Inclusion) initiatives.
